

CSCI 112

Programming with C

LECTURE: 4

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OUTLINE:

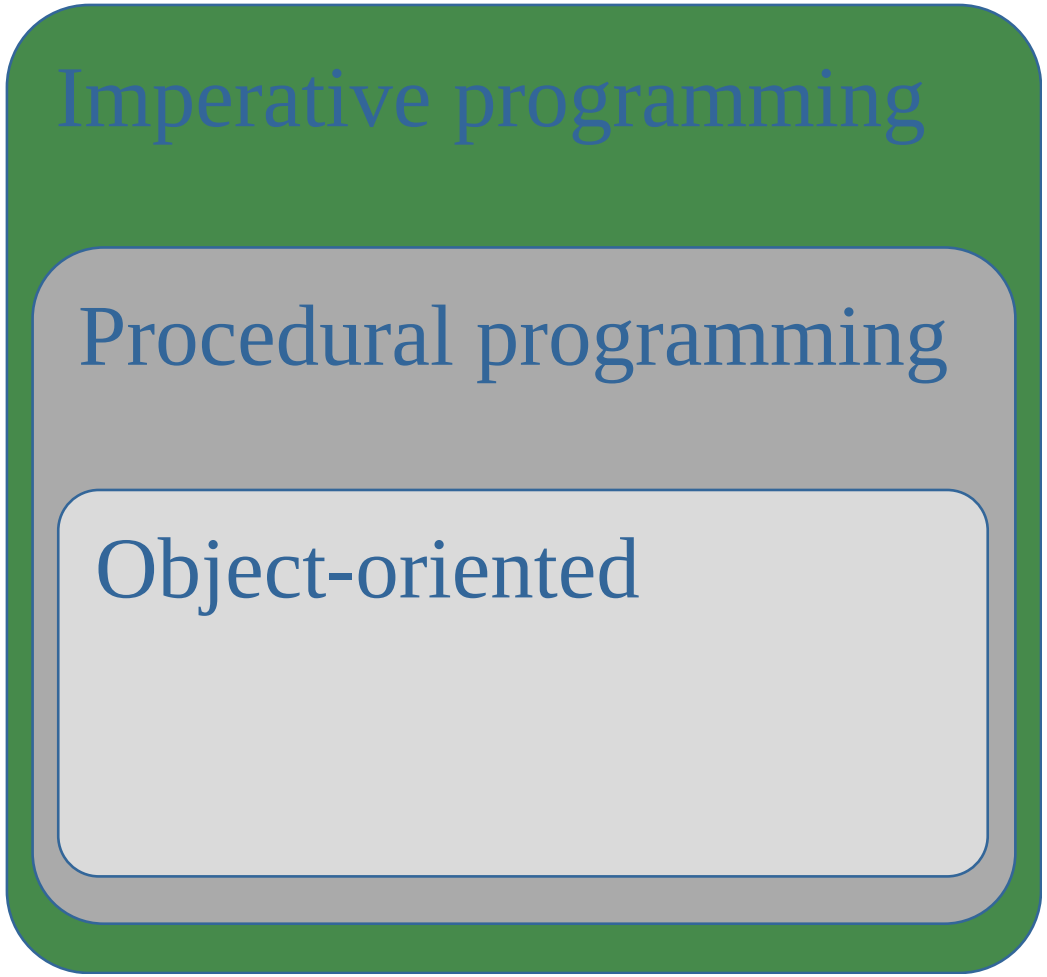
- Statements & expressions
- Block
- Accessing Array Elements
- Logical vs Bitwise Operators
- Sizeof()
- const
- Limits & float
- Input / Output

STATEMENTS

& expressions

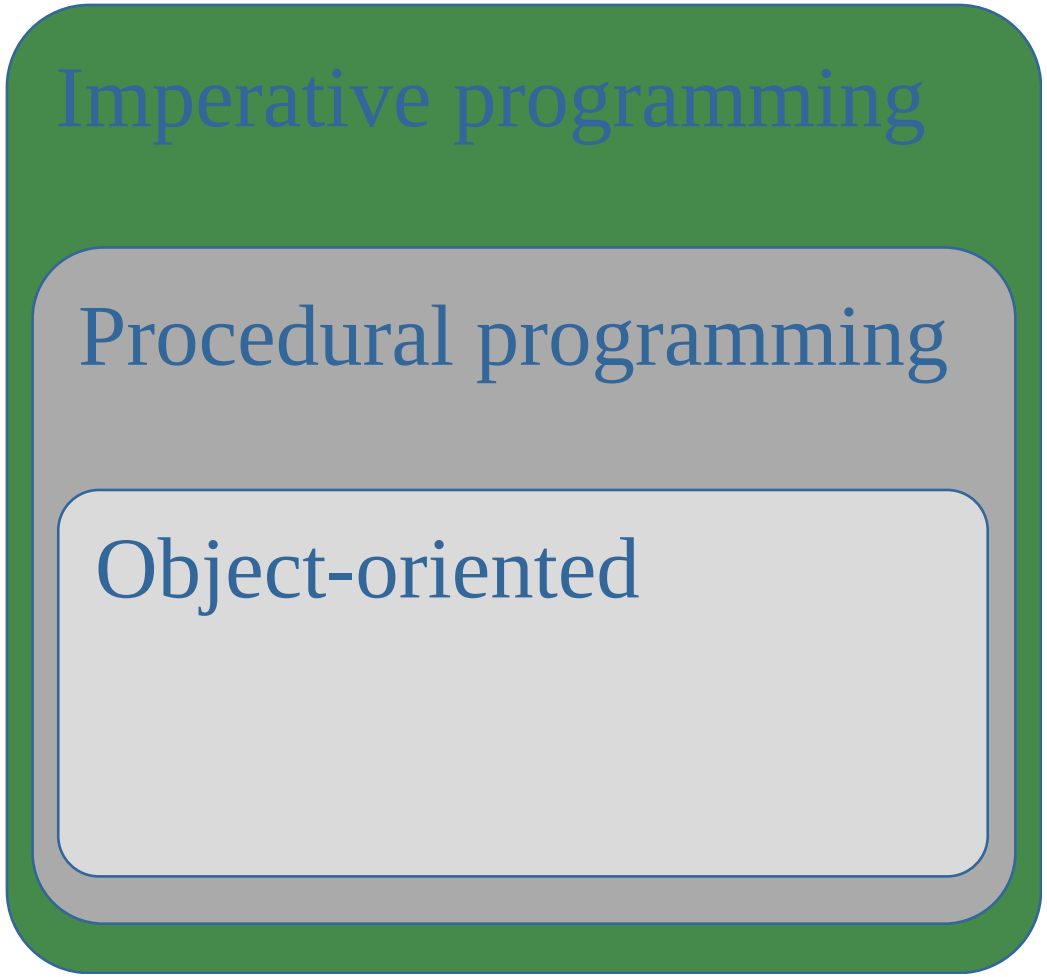
IMPERATIVE PROGRAMMING

- **Imperative programming** is a programming paradigm that uses s**tatements to change a program's state**.
- It is like giving the computer instructions step by step on what to do.



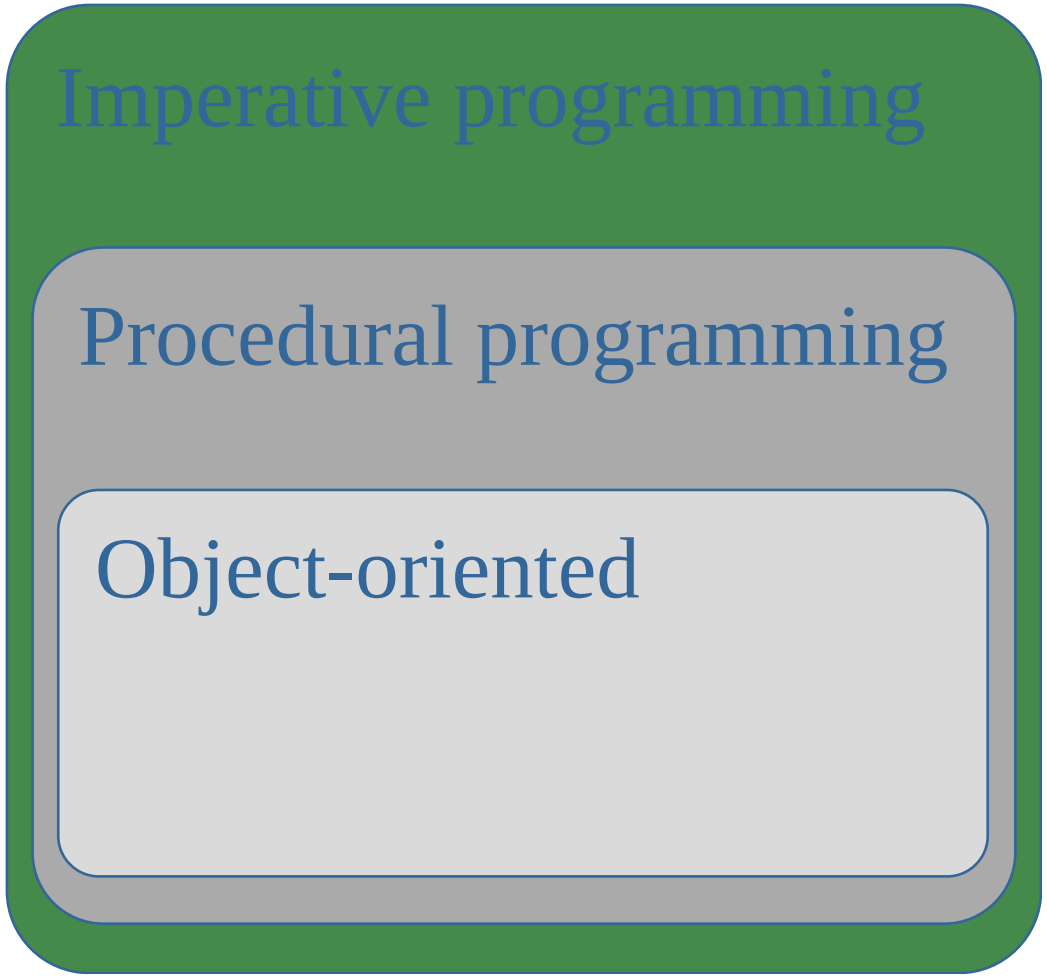
PROCEDURAL PROGRAMMING

- **Procedural programming** is a type of imperative programming that breaks down a program into smaller, independent procedures (functions).
- This makes the program more organized and easier to maintain.



OBJECT-ORIENTED PROGRAMMING

- **Object-oriented programming** further extends the concepts of procedural programming by organizing code into objects that encapsulate data and behavior.



STATEMENT & EXPRESSION

- In computer science, a **statement** is a complete instruction that can modify the state of a program, such as assigning a value to a variable, calling a function, or jumping to a different part of the code. **Statements do not produce a value on their own.**
- An **expression**, on the other hand, is a combination of variables, constants, functions, and operators that evaluates to a single value (**The absence of a returned value from a function is also a result**). Expressions can be used within statements to provide values for operations or to control the flow of the program. However, **expressions themselves do not directly modify the state of the program.**

NULL / EMPTY STATEMENT;

```
int main()
{
        ;                               /*  Empty statement    */
        printf("Hello world!\n");                   /*  Second statement   */
        printf("Hello"); printf(" world!\n");               /*  3rd & 4th statement in one line */
}
```

- A single semicolon `;` in C language is a statement called a `null` statement or empty statement.
- A semicolons `;` are used to terminate statements

CONCLUSIONS

- *In C, a code line is not the same as a statement*
- *An entire program can be written in a single line*
- *As the compiler separates statements with semicolons* `;` *and not newlines*
- *Therefore, a single line of text (with a newline character) can contain multiple statements.*

BLOCK

BLOCK & COMPLEX(COMPOUND) STATEMENTS

- **Block** can be:
  - substitute for simple statement,
  - defined within a block (Nested blocks),
  - empty `{}`.
- Variables can be declared inside
- Variables declared inside a block with the same name hide access to variables outside that block
- No semicolon at end of block
- After a code block ends, all variables declared within that block cease to exist and are no longer accessible in the rest of the program.

BLOCK & COMPLEX(COMPOUND) STATEMENTS - EXAMPLE

```
int main()
{
    char* text = "Hello world\n";
    printf(text);                          // Hello world
    {
      char* text = "Bye world\n";
      printf(text);                        // Bye world
      {
        char* text = "Aloha world\n";
        printf(text);                      // Aloha world
      }
    }
    printf(text);                          // Hello world
}
```

SUMMARY

- A **statement** can modify the state of a program, such as assigning a value to a variable, calling a function, or jumping to a different part of the code.
- Statements do not produce a value on their own.
- An **expression** is a combination of variables, constants, functions, and operators that evaluates to a single value.
- Expressions themselves do not directly modify the state of the program.

EXAMPLE

```
int main()
{
        int x, y, _;
        _ = x = y = 5;

        _ = x = y = 5;
        _ = x = (5);
        _ = x = 5;
        _ = (5);
        _ = 5;
        (5)
}
```

CONSEQUENCES

- In the C language, **every** assignment after assigning a value to a variable (pointer, label), since it is also an expression, will generate a value again, which will be ignored by the compiler at the end.
- We can use the underscore `_` discard in C, but you need to declare the variable beforehand. It's a great mechanism that improves the readability of the programmer's intent.

ACCESSING

Array Elements

ACCESSING ARRAY ELEMENTS 1/2

```c
#include <stdio.h>
int main()
{
    int tab[3];       /* Declare an array of 3 integers */
    /* Store values into the array */
    tab[0] = 5;       /* put 5 into the first element */
    tab[1] = 10;      /* put 10 into the second element */
    tab[2] = 15;      /* put 15 into the third element */
    /* Retrieve values from the array */
    int x = tab[0];   /* read the first element into variable x */
    int y = tab[1];   /* read the second element into variable y */
    /* Display the values */
    printf("x = %d\n", tab[0]);   // prints 5
    printf("y = %d\n", tab[1]);   // prints 10
    return 0;
}
```

ACCESSING ARRAY ELEMENTS 2/2

- `tab[index]` is used both to read and write values.
- Array indices in C start from 0.
- Make sure the index is within the declared size.
- Each element in the array is stored contiguously in memory.

```
struct Point { int x; int y; }; int main()
{struct Point point = {1,2}, *ppoint = &point; int arr[] = {1,2}; int x = 5, y =-6; int * z; float f = 3.0f; /*code*/}
```

Priority	Operator	Description	Associativity	Example	Result
1	**(), []**	**Parentheses; Array subscript**	**Left-to-right**	**arr[0] * (x + y)**	**1**
	.	Structure and union member access		point.x	**1**
	->	Structure and union member access through pointer		ppoint->x	**1**
2	++, --	Prefix & postfix increment and decrement	Right-to-left	++x; x--; x;	**6, 6, 5**
	+, -, !, ~	(Unary) plus and minus; Logical NOT and bitwise NOT		y =-y; y =+y; !x; ~x	**6, -6,0, -6**
	*, & , &&	Indirection (dereference); Address-of; Address-of labels		z = &x; *z;	**6422276; 5**
	(type), sizeof	Cast, Size-of		(int)3.0f; sizeof(x);	**3, 4**
3	*, /, %	Multiplication, division, and remainder	Left-to-right	6/2 % 2	**1**
4	+, -	Addition and subtraction		1 + 2; 3 - 1	**3, 2**
5	<<, >>	Bitwise left shift and right shift		4 << 1; 4 >> 2	**8, 1**
6	<, <=, >, >=	For relational operators <, > and ≤, ≥ respectively		x<y; x<=y; x>y; x>=y	**1, 1, 0 ,0**
7	==, !=	For relational = and ≠ respectively		x == y, x != 1	**1, 0**
8	&	Bitwise AND		7 & 3	**3**
9	^	Bitwise XOR (exclusive or)		255 ^ 0	**255**
10	|	Bitwise OR (inclusive or)		7 | 3	**7**
11	&&	Logical AND		1 && 0	**0**
12	||	Logical OR		1 || 0	**1**
13	?:	Ternary conditional	Right-to-left	x = (x > y) ? y : x;	**-6**
14	=	Simple assignment		x = y;	**-6**
	+=, -=, *=, /=, %=	Assignment by sum, difference, product, quotient, remainder		x+=1; x-=1; //etc.	**6, 5**
	<<=, >>=, &=, ^=, |=	Assignment by bitwise left shift, right shift, AND, XOR, OR		3<<=1, 8>>=2 //etc.	**6, 2**
15	,	Comma	Left-to-right	x = 3, y = 1;	**3**

LOGICAL

vs Bitwise Operators

LOGICAL VS BITWISE OPERATORS

In C, Boolean logic can be used in two ways:

Bitwise operations on variables:

- the result is a numeric value obtained by applying the corresponding logic gate to each bit of the variables involved. The exception is the bitwise NOT `~`, which inverts all bits.

Evaluating the logical value of an expression:

- remember that any non-zero value (including negative numbers) is treated as `true` (1), and zero (or NULL) is `false` (0). Essentially, the numeric value of each variable is first cast to `0` or `1`, and then the logical expression is evaluated.

BITWISE OPERATORS

```
unsigned int x = 6;          /* 00000000 00000000 00000000 0000 0110 (binary) */
unsigned int y = 3;          /* 00000000 00000000 00000000 0000 0011 (binary) */

/* AND & */
printf("%u\n", x & y);       /* (00000000 00000000 00000000 0000 0010) */          //2
/* OR  | */
printf("%u\n", x | y);       /* (00000000 00000000 00000000 0000 0111) */          //7
/* NOT ~ (flips all bits) */
printf("%u\n", ~x);          /* (11111111 11111111 11111111 1111 1001) */          //4 294 967 289
/* XOR ^ */
printf("%u\n", x ^ y);       /* (00000000 00000000 00000000 0000 0101) */          //5
```

LOGICAL OPERATORS

```
unsigned int x = 6;             /* 00000000 00000000 00000000 0000 0110 (binary) */
unsigned int y = 3;             /* 00000000 00000000 00000000 0000 0011 (binary) */

/* AND && */
printf("%d\n", x && y);                 /* (00000000 00000000 00000000 0000 0001) */     // 1 (true)
/* OR  || */
printf("%d\n", x || y);         /* (00000000 00000000 00000000 0000 0001) */     // 1 (true)
/* NOT ! */
printf("%d\n", !x);             /* (00000000 00000000 00000000 0000 0000) */     // 0 (false)
/* XOR doesn't exist but =  (a || b) && !(a && b); */
printf("%d\n", (x || y) && !(x && y)); /* (00000000 00000000 00000000 0000 0000) */         // 0 (false)

printf("%d\n", -1 && -2);               /* (00000000 00000000 00000000 0000 0001) */     // 1 (true)
```

SIZEOF()

SUMMARY OF MEMORY SIZE OF DATA TYPES

Type	Memory size in bytes / bits
`char`	01 Bytes / 08 bits
`bool` *	01 Bytes / 08 bits
`short int`	02 Bytes / 16 bits
`long int`	04 Bytes / 32 bits
`float`	04 Bytes / 32 bits
`double`	08 Bytes / 64 bits
`long double`	12 Bytes / 96 bits
pointer (`char` *, `short` *, `long` *, `int` *, `float` *, `double` *, `long double` *, `void` *, …)	04 Bytes / 32 bits
user defined (`struct` , `union` , etc.)	complex

*from C99

OPERATOR SIZEOF()

The sizeof operator returns the size of a **type**(requires parentheses) or **variable** in bytes.

- It is a **unary operator**, just like a cast, and has the same precedence level.
- It is easy to mistake sizeof for a function, but it is not — it is evaluated at compile time.

```
long long x, short y, float z;
printf("%d\n", sizeof(x));          // 8
printf("%d\n", sizeof x );          // 8
printf("%d\n", sizeof(long long));      // 8
printf("%d\n", sizeof(y));          // 2
printf("%d\n", sizeof y );          // 2
printf("%d\n", sizeof(short) );         // 2
printf("%d\n", sizeof(z));          // 4
printf("%d\n", sizeof z );          // 4
printf("%d\n", sizeof(float) );         // 4
```

CONST

NAMED CONSTANTS IN C

When you see names written in **ALL CAPS** in C code, they represent **constants** — values that do not change during program execution.

The `const` Keyword

You can create a constant by adding `const` before the type:

```
const int MAX_STUDENTS = 30;
const double PI = 3.14159;
```

- `const` **tells the compiler:** this value must never be changed.
- **If you try to modify it, the compiler will report an error.**

NAMING CONVENTION

Constants are written in `ALL_CAPS_WITH_UNDERSCORES`.

```
const int MAX_SIZE   = 100;
const int MIN_AGE    = 18;
const double GRAVITY = 9.81;
```

This is a widely used convention in C — it makes constants easy to spot and distinguish from regular variables at a glance.

Why Does This Matter?

When you see names like INT_MAX or FLT_MAX in C, they follow exactly the same convention — they are constants that represent the largest value an int or float can hold.

LIMITS & FLOAT

LIMITS.H & FLOAT.H

- To access constants defining the minimum and maximum values of integer types in C,
  - include the header `<limits.h>`,
  - For floating-point types, include `float.h`.
- Important: `FLT_MIN` (and other floating-point *min* constants) does not represent the most negative value. It is the smallest positive non-zero value that can be represented.

If you want the actual minimum (most negative) value, use `-FLT_MAX`.

LIMITS.H

```
printf("CHAR_MIN          = %d\n", CHAR_MIN);          /* char */
printf("CHAR_MAX           = %d\n", CHAR_MAX);
printf("UCHAR_MAX          = %u\n", UCHAR_MAX);

printf("SHRT_MIN         = %d\n", SHRT_MIN);          /* short */
printf("SHRT_MAX          = %d\n", SHRT_MAX);
printf("USHRT_MAX         = %u\n", USHRT_MAX);
```

```
Result:
---
CHAR_MIN          = -128
CHAR_MAX           = 127
UCHAR_MAX          = 255
SHRT_MIN         = -32768
SHRT_MAX          = 32767
USHRT_MAX         = 65535
```

FLOAT.H

```
printf("INT_MIN          = %d\n", INT_MIN);        /* int */
printf("INT_MAX           = %d\n", INT_MAX);
printf("UINT_MAX          = %u\n", UINT_MAX);

printf("FLT_SMALLEST      = %e\n", FLT_MIN);        /* float */
printf("FLT_MIN           = %f\n", -FLT_MAX);
printf("FLT_MAX           = %f\n", FLT_MAX);
printf("FLT_MAX           = %e\n", FLT_MAX);
```

```
INT_MIN          = -2147483648
INT_MAX           = 2147483647
UINT_MAX          = 4294967295
FLT_SMALLEST      = 1.175494e-038
FLT_MIN          = -340282346638528859811704183484516925440.000000
FLT_MAX           = 340282346638528859811704183484516925440.000000
FLT_MAX           = 3.402823e+038
```

INPUT / OUTPUT

INPUT / OUTPUT

Input and output functions in C are not available by default — you need to include the `<stdio.h>` library. The most basic functions include:

- `getchar()` – reads a single character from the keyboard,
- `putchar()` – displays a single ASCII character on the screen,
- `printf()` – prints a formatted string to the screen,
- `scanf()` – reads a formatted string from the keyboard.

INPUT / OUTPUT

In C, it **is not** easy to explain how these functions work internally without more advanced knowledge. Therefore, we will start by using only the output functions:

⚒ `putchar()` and `printf()`.

To better understand how input works, we will first use the **getche()** function from the Windows-only `conio.h` library. After that, we will move on to `getchar()` and finally to `scanf()`.

`Getche() - (get[ ]char[acter with echo])`

FUNDAMENTAL FUNCTIONS FOR INPUT AND OUTPUT

- Data Output for Screen:
  - `putchar` * - (put[ ]char[acter]): Displays a single character on the screen.
  - printf - (print[ ]f[ormatted]): Displays a formatted string of characters.
- Data Input from Keyboard:
  - `getchar` * - (get[ ]char[acter]): Retrieves a single character from the keyboard.
  - `scanf` - (scan[ ]f[ormatted): Reads a formatted string of characters from the keyboard.
- C doesn't handle input/output on its own. You need a library called `<stdio.h>`.
- *You can use putchar and getchar like normal functions, but they are not standard C functions but rather preprocessor macros.

PUTCHAR() & GETCHE() WITHOUT \N

```
int main()
{
    char character;
    character = getche();   /* Read a single key from the keyboard and immediately display it (echo) */
    putchar(character);     /* Display the value of 'character' using putchar (prints the same key again) */

    character = getche();   /* Read another single key (char is enough, automatic casting occurs) */
    putchar(character);     /* Display the second key pressed */
} // Result: aaBB
```

`int putchar(int);`

- putchar(c) puts the character c on the standard output. It returns the character printed or EOF on error(-1).

  `int getche(void);`

- returns the next character from standard input. It returns `EOF` (end of file) on error.

PUTCHAR() & GETCHE() WITH \N

```
int main()
{
    char character;
    character = getche();          /* Read a single key from the keyboard and immediately display it (echo) */
    putchar('\n');             /* Display the CRLF using putchar to move to the next line for clarity */
    putchar(character);    /* Display the value of 'character' using putchar (prints the same key again) */
    putchar('\n');             /* Display the CRLF using putchar to move to the next line for clarity */

    character = getche();          /* Read another single key (char is enough, automatic casting occurs) */
    putchar('\n');             /* Display the CRLF using putchar to move to the next line for clarity */
    putchar(character);    /* Display the second key pressed */
    putchar('\n');             /* Display the CRLF using putchar to move to the next line for clarity */
}
```

```
a
a
B
B
```

CHAR ISSUE 1/2

- By default, `char` in C is `signed`, which means its range is -128 to 127 (two's complement) instead of 0–255.
- When you write: `char var = 255;` or something greater than 255+x;
- you need to know that 255 is by default an int (4 bytes).

CHAR ISSUE 2/2

- An automatic cast occurs here: only the lowest 8 bits are kept, and the most significant bit (**MSB**), which determines the sign, is overwritten. As a result, `printf()` will show `-1` because that is how these 8 bits are interpreted in signed form.
- This is an example of overflow and truncation, causing loss of information. This is also why `getche()` / `getchar()` returns an int: it can hold all possible `char` values plus the special `EOF` value `-1`. Using int ensures we can reliably distinguish between a real `char` with all bits set and an intentional `EOF` signal.

BASICS OF INT PRINTF (CHAR FORMAT[], ARG1, ARG2 ,...);

- The format requires a string enclosed in double quotes (`"<string>"`).
- If we want to display the contents of our variables, such as `int` types, we must use the `%` symbol followed by the type. This allows displaying a value from memory interpreted as the given type and after we must provide the name of the variable that will be read. Each instance of `%<type>` will allow us to display the contents of one variable.

```
int main()
{
  int x = 5;
  int y = 7;
  printf ("%s\n", "Hello world\n");                    // Hello world
  printf("%d\n", x);                                   // 5
  printf("Value of x = %d, and value of y = %d\n", x, y);    // Value of x = 5, and value of y = 7
  printf("%d %d\n", x, y, x, x);                       // 5 7
}
```

THANK

You